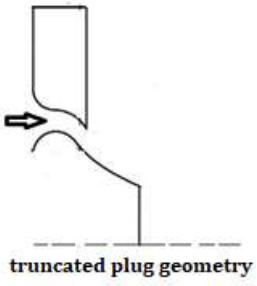
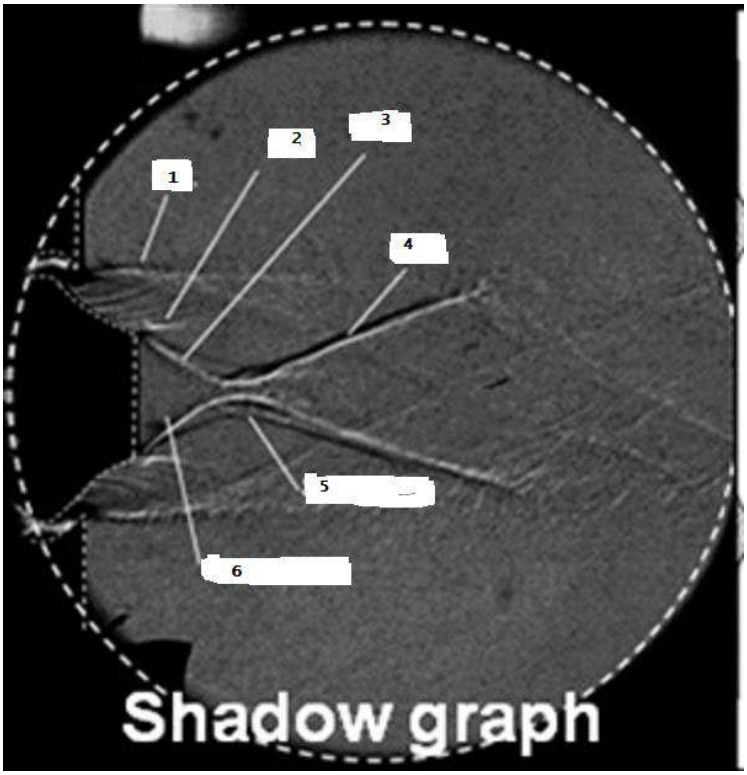
25.01.2023 - Gas Dynamics - MSAR - B. Favini

1. A supersonic steady two-dimensional uniform stream at mach number **M = 1.3** is flowing along a flat plate where impinges an oblique shock. Define the relative pressure-deflection polar and determine the different possible oblique shock reflections (assume ideal gas model). Describe the identified different reflections in the physical (x,y) plane. **(score 13/30)**

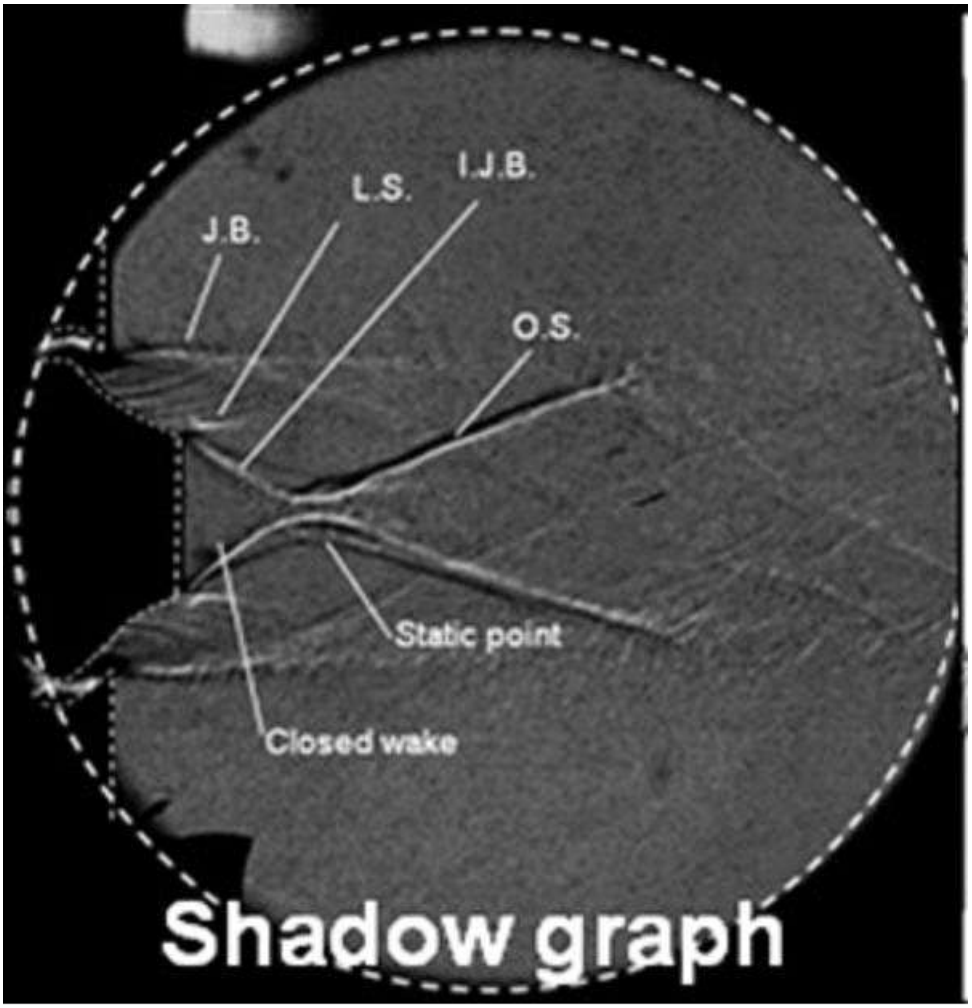
2. A piston located at the left boundary of a very long pipe moves to the right with parabolic law $\mathbf{x = t^2}$ up to **t = 1** and after this time with linear law **x = 2t - 1**. The gas inside the pipe is at rest with pressure **p = 1** and density $\boldsymbol{\rho = \gamma}$. Determine the evolution of the field and its dependence on the gas molecular weight (properties $\boldsymbol{\gamma}$ and **R)**. Describe the flowfield on the space-time and pressure-velocity planes. **(score 13/30)**

3. Identify waves and regions numbered in the visualization of the supersonic steady flow generated by a two-dimensional truncated plug nozzle. **(score 4/30)**





3. Plug flow shadowgraph



nozzle pressure ratios NPR = 30

J.B.:Jet boundary, I.J.B.: Inner Jet boundary, L.S.: Lip Shock, O.S.: Oblique Shock